

Descriptor: *AEGIS™ Threat Matrix for Agentic AI Systems (ATX-1)*

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ABSTRACT ATX-1 (AEGIS Threat Matrix) is a structured adversarial knowledge base that catalogs failure modes of autonomous AI agents operating without governance constraints. Unlike existing threat frameworks—MITRE ATT&CK for human adversaries and MITRE ATLAS for adversaries targeting AI systems—ATX-1 addresses a distinct threat class: AI agents that act outside their governance boundaries not through external compromise, but through their own structural capability without corresponding authority. ATX-1 is grounded in a fundamental distinction: capability—what an agent can do—is not authority—what an agent may do. The ~~five~~^{four} structural root causes identified in the dataset (missing authority verification, missing consequence modeling, missing behavioral boundaries, [missing state integrity protection](#), and [missing environment model](#)~~and missing state integrity protection~~) trace all [29 techniques \(and 29 sub-techniques\)](#)~~20 techniques~~ to architectural gaps that governance frameworks can address. The dataset defines [10 tactics \(TA001–TA010\)](#)~~9 tactics (TA001–TA009)~~, [29 techniques with 29 sub-techniques distributed non-sequentially across tactics \(T1001–T10004 with sub-techniques T####.###\)](#)~~20 techniques distributed non-sequentially across tactics (T1001–T9001)~~, and [29 mitigations](#)~~20 mitigations~~, each grounded in empirical observations from the *Agents of Chaos* study, which documented 11 failure modes across live agentic deployments. ATX-1 is published in STIX 2.1 format for compatibility with threat intelligence platforms, with supplementary structured JSON files providing technique metadata, regulatory cross-references to the NIST AI Risk Management Framework, EU AI Act, and OWASP Top 10 for LLM Applications, and a JSON Schema for validation. The dataset satisfies FAIR principles through three permanent repositories—IEEE DataPort, Zenodo, and a live machine-readable API—under the Apache 2.0 license.

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DATA DOI/PID [10.5281/zenodo.20171712](https://doi.org/10.5281/zenodo.20171712) (ATX-1 v2.3)~~10.21227/f87b-1d57 (ATX-1 v1.0)~~

DATA TYPE/LOCATION Threat Taxonomy; Online (IEEE DataPort, Zenodo, aegis-governance.com)

INDEX TERMS Agentic AI, AI governance, autonomous agents, constitutional enforcement, EU AI Act, NIST AI RMF, OWASP LLM, STIX 2.1, threat taxonomy

BACKGROUND

Autonomous AI agents—systems that execute real-world actions against operational infrastructure including API calls, file system operations, database queries, and inter-service requests—have grown significantly in production deployments during the twelve months preceding this publication [?]. This growth has outpaced the development of governance infrastructure suited to the action layer, creating a systematic gap between what agents can do and what they are authorized to do.

Existing threat taxonomy frameworks do not address this gap. MITRE ATT&CK [?] catalogs adversarial tactics and techniques used by human threat actors against computer systems. MITRE ATLAS [?] catalogs adversarial tactics used against AI and machine learning systems. Both frameworks assume a clear attacker-target separation: an external adversary acts against a system. ATX-1 addresses the scenario where the AI agent itself is the threat source—not through compromise or malicious intent, but through capability without governance constraint.

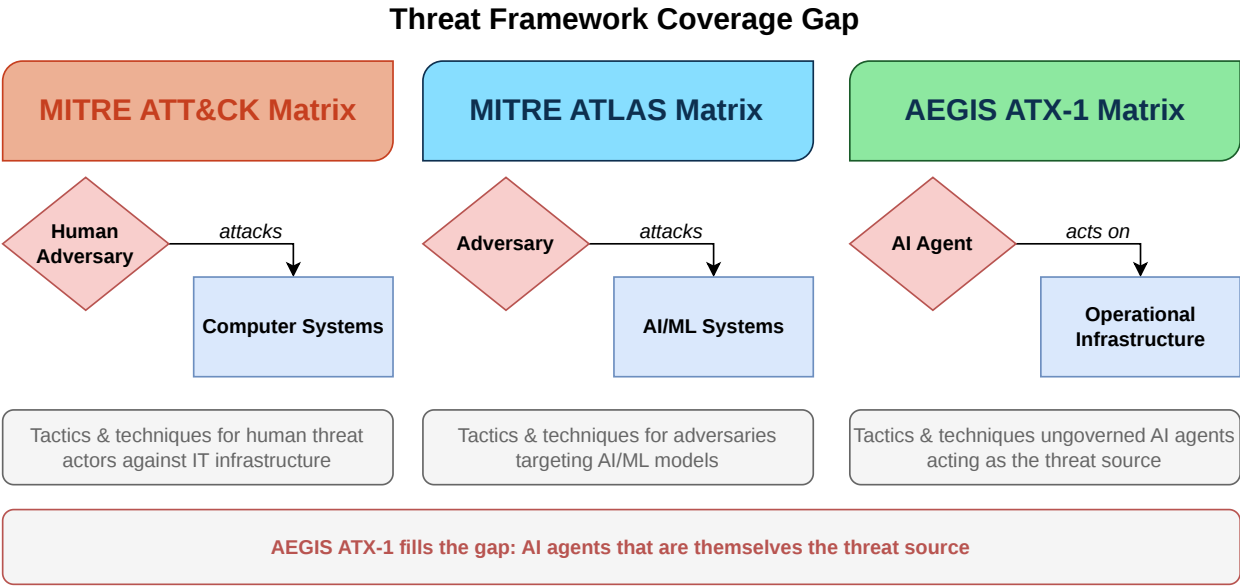


FIG. 1. Threat framework coverage gap. MITRE ATT&CK catalogs human adversaries attacking computer systems. MITRE ATLAS catalogs adversaries attacking AI/ML systems. ATX-1 addresses the uncovered scenario: AI agents that act on operational infrastructure without governance constraint. The distinction between “attacks” and “acts on” reflects the structural difference—ATX-1 agents are not adversaries but ungoverned systems.

The distinction is structural. ATT&CK and ATLAS model adversaries who attack systems. ATX-1 models agents that are themselves the threat source—systems that delete data not because an attacker told them to, but because no architectural mechanism verified that the instruction source held the authority to request deletion. The *Agents of Chaos* study (Shapira et al., 2026) [?] documented this failure pattern systematically, identifying 11 distinct failure modes across live agentic deployments involving 20 researchers over two weeks.

Multiple independent research groups have converged on the same finding. Arora et al. (2025) demonstrated that common multi-agent design patterns—including plan construction strategies, inter-agent context sharing, and fallback behaviors—carry significant vulnerabilities that emerge from architecture and interaction, not solely from adversarial prompts [?]. Ko et al. (2025) identified seven categories of security challenges in cross-domain multi-agent systems, establishing that “a benign agent in isolation may leak secrets or violate policy” when interacting with other agents across trust boundaries [?]. Reid et al. (2025) formalized six failure classes for governed multi-agent systems—including cascading failures, conformity bias, and mixed-motive dynamics—concluding that “a collection of safe agents does not guarantee a safe collection” [?]. These studies independently corroborate the core premise of ATX-1: that agent-level safety does not ensure system-level safety, and that a distinct threat taxonomy is required for the agent-to-infrastructure action layer.

ATX-1 was developed as part of the AEGIS (Architectural Enforcement and Governance of Intelligent Systems) project [?], an open governance architecture that applies reference monitor principles [?] to enforce deterministic constitutional policy at the agent action boundary. The taxonomy directly informs the AEGIS threat model (ATM-1) and maps each technique to specific constitutional articles and governance protocol mechanisms that mitigate it.

No comparable machine-readable dataset exists for agentic AI threat modeling. The OWASP Top 10 for LLM Applications [?] provides a high-level risk categorization but does not define individual techniques, map mitigations to architectural mechanisms, or provide STIX-compatible data for integration with threat intelligence platforms. ATX-1 fills this gap.

COLLECTION METHODS AND DESIGN

Empirical Foundation

ATX-1 techniques are derived from the *Agents of Chaos* study [?], which conducted a controlled evaluation of autonomous AI agent robustness. The study deployed AI agents in realistic operational scenarios and systematically documented failure modes. Twenty researchers participated over a two-week period, producing 11 case studies (CS1–CS11) documenting specific failure instances.

Taxonomy Construction

The ATX-1 taxonomy was constructed through a four-stage process:

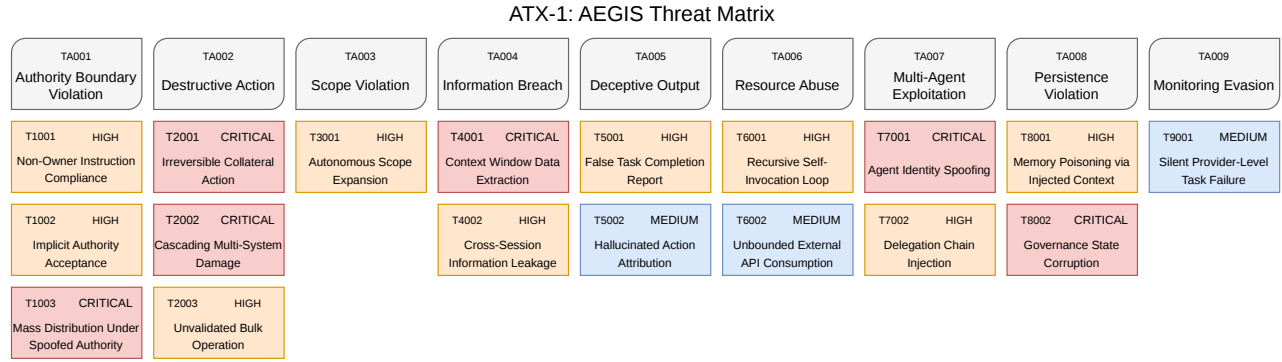


FIG. 2. ATX-1 threat matrix. Ten tactic columns (TA001–TA010) contain 29 techniques with 29 sub-techniques. Techniques are color-coded by severity: critical (red), high (orange), medium (blue). Technique identifiers follow a non-sequential scheme where the first two digits encode the parent tactic (e.g., T1001–T1003 belong to TA001); sub-techniques use the form T###.###. v2.3 removes the severity field to align with MITRE ATT&CK and ATLAS conventions, which leave contextual scoring to the consumer.

1. **Case study analysis.** Each of the 11 *Agents of Chaos* case studies was analyzed to extract the specific failure mechanism, the conditions under which it occurred, and the consequences observed.

2. **Root cause identification.** Five structural root causes were identified that enable agentic failures:

- **RC1: Missing Authority Verification**—No formal model of principals, delegation chains, or authority levels exists for the agent to verify instruction sources.
- **RC2: Missing Consequence Modeling**—The agent has no representation of the impact, reversibility, or scope of its actions before executing them.
- **RC3: Missing Behavioral Boundaries**—No formal boundary defines the scope of the agent’s task, allowing autonomous expansion beyond original instructions.
- **RC4: Missing State Integrity Protection**—Instructions and data commingle in the agent’s context, making injection structurally possible.
- **RC5: Missing Environment Model**—The governance layer operates on an abstraction of the execution environment but lacks a complete model of the environment’s actual capabilities. Actions evaluated as one type may produce entirely different effects depending on how the environment interprets them.

3. **Tactic and technique classification.** Failure modes were grouped into 10 tactics (categories of ungoverned behavior) and 29 techniques with 29 sub-techniques (specific failure mechanisms). Each technique was assigned a severity rating (critical, high, medium, or low), linked to its root cause(s) using the RC1–RC5 labels in the dataset’s `root_cause` field, and mapped to the originating case study (where v1.0 derivation applies; v2.1 techniques in tactic TA010 derive from RFC-0006 adversarial testing of the AEGIS Claude Code plugin).

4. **Mitigation mapping.** Each technique was mapped to specific AEGIS constitutional articles and AGP-1 governance protocol mechanisms that architecturally prevent the failure mode. Regulatory cross-references to NIST AI RMF functions, EU AI Act articles, and OWASP LLM Top 10 categories were added.

Construction Reproducibility

The four-stage process above involves discretionary judgments at each stage. To make these judgments reproducible by independent coders, the following rules and criteria were applied across all 11 source case studies of v1.0, the four RFC-0006 adversarial-testing rounds that produced v2.1’s TA010 additions, and the bypass-method observations that produced v2.2’s sub-techniques.

Case-to-root-cause coding. Each observed failure was coded for the structural factor that enabled it. The coding rules are:

- Code as **RC1** when the agent acted on an instruction whose source authority was not architecturally verified.
- Code as **RC2** when the agent performed an action without architectural representation of impact, reversibility, or scope.
- Code as **RC3** when the agent expanded scope beyond declared task boundaries with no boundary check.
- Code as **RC4** when failure proceeded through commingled instructions and data (prompt injection, indirect injection, or context poisoning).
- Code as **RC5** when the agent’s action took an environment-interpreted path that the governance abstraction did not model.

Multiple root causes per technique are permitted; the dataset’s `root_cause` field accepts a comma-separated set. Inter-coder reliability was not measured (single-author coding); the dataset’s `agents_of_chaos_case` field provides direct case-to-technique traceability, enabling independent recoding by future researchers.

Case-to-technique abstraction. A failure mode is divided into separate techniques when (i) the mechanism differs (different conditions, different invariants violated), (ii) the architectural mitigation differs (different governance primitive blocks it), or (iii) the threat surface differs (different action category or capability). Multiple cases collapse into a single technique when mechanism, mitigation, and threat surface all align; only surface details (target identifier, tooling, payload) differ. In v1.0 this rule produced a 11-cases-to-20-techniques ratio: some cases (e.g., AoC CS#1 "non-owner instruction compliance") yielded multiple techniques because the same observation surfaced two distinct mechanisms, and some techniques (e.g., T2001 "Perform Irreversible Destructive Action") aggregate multiple cases that converged on the same mechanism.

Sub-technique introduction. A sub-technique (introduced in v2.2 under IDs of the form T####.###) is created when a specific bypass method of the parent technique is sufficiently distinct in mechanism to warrant its own identifier, AND the bypass mechanism cannot be addressed by the parent technique's mitigation alone. v2.3 carries 29 sub-techniques, all under T9002 (Obfuscate Action Traceability) and the four TA010 parent techniques (T10001–T10004). Sub-techniques inherit the parent's tactic, root cause, and primary mitigation, but carry their own mechanism descriptions and may add specialized mitigations.

Tactic-boundary rules. A tactic is a category of ungoverned outcome—a class of agentic results, not a class of mechanisms. Tactic boundaries are determined by three factors in priority order: (i) the agent's effective goal (the kind of result the failure produces), (ii) the architectural layer affected (authority verification, scope control, integrity protection, state integrity, etc.), and (iii) the dominant mitigation pattern (which AEGIS primitive most directly prevents the failure). Tactic membership for a technique is decided by which of these three factors most cleanly classifies it. Where a technique could reasonably belong to two tactics, the primary tactic is the one whose mitigation pattern aligns most directly; the secondary association is recorded as a relationship in the STIX bundle rather than duplicating the technique entry.

End-to-end processing flow. The four-stage process maps inputs to outputs as: (1) raw case study or adversarial-test observation → failure-mechanism description; (2) failure-mechanism description → RC1–RC5 assignment per the coding rules above; (3) root cause + mechanism + threat surface → technique identifier (T#### or T####.###) and tactic assignment per the boundary rules above; (4) technique → AEGIS constitutional article + AGP-1 mechanism + regulatory cross-references. The complete pipeline, including the source case studies, technique records, root cause assignments, and mitigations, is published in the dataset's `atx-1-techniques.json`

(full records) and `atx-1-version-mapping.json` (per-version derivation provenance) for independent verification.

Machine-Readable Encoding

The taxonomy was encoded in four machine-readable formats:

STIX 2.1 Bundle. The complete taxonomy is represented as a STIX 2.1 Bundle [?] containing: 1 Identity object (AEGIS Initiative), 109 x-mitre-tactic objects, 5820 attack-pattern objects (29 parent techniques and 29 sub-techniques, each with kill chain phases and external references~~one per technique with kill chain phases and external references~~), 2920 course-of-action objects (mitigations), and 106 relationship objects encoding tactic-to-technique (uses), mitigation-to-technique (mitigates), parent-to-sub-technique (subtechnique-of), and technique-to-OWASP (related-to) edges~~54 relationship objects: 20 tactic-to-technique (uses), 20 mitigation-to-technique (mitigates), and 14 technique-to-OWASP (related-to)~~. The STIX format was chosen for compatibility with existing Security Information and Event Management (SIEM) platforms and threat intelligence tools.

Technique Database (JSON). A flat JSON array containing all 58 entries (29 parent techniques and 29 sub-techniques)~~20 techniques~~ with structured metadata: identifier, name, parent tactic, description, ~~severity~~, root cause, case study references, OWASP mapping, parent-technique and sub-technique relations, and AEGIS mitigation details.

TABLE 1. Technique Record Structure

Field	Type	Description
id	string	Technique ID (T1001–T10004 with sub-techniques) (T1001–T9001)
name	string	Human-readable name
tactic	string	Parent tactic (TA001–TA010) (TA001–TA009)
tactic_name	string	Parent tactic name
description	string	Failure mode description
severity	enum string	critical/high/medium/low
<u>parent_technique</u>	<u>string</u>	<u>Parent technique ID if sub-tech</u>
<u>sub_techniques</u>	<u>array</u>	<u>Sub-technique IDs if parent</u>
root_cause	string	Root cause (RC1–RC5) (RC1–RC4)
agents_of_chaos_case	array	Case study numbers [?]
owasp_mapping	array	OWASP LLM Top 10
aegis_mitigation	object	Article, AGP mechanism

Regulatory Cross-Reference (JSON). A structured mapping of each technique to NIST AI RMF [?] functions and categories, EU AI Act [?] articles, OWASP LLM Top 10 categories, and ATM-1 threat scenarios.

JSON Schema. A JSON Schema (Draft 2020-12) defining the structure and validation rules for ATX-1 technique definitions, enabling automated validation of technique entries.

VALIDATION AND QUALITY

Empirical Grounding

Every technique in ATX-1 is traceable to at least one case study from the *Agents of Chaos* study [?]. The mapping between techniques and case studies is explicit in the dataset: each technique record includes an `agents_of_chaos_case` field containing the case study number(s) that demonstrated the behavior. No technique was included without empirical evidence of the failure mode occurring in a live agentic deployment.

Schema Validation

All technique entries in the technique database conform to the ATX-1 JSON Schema (`atx-technique.schema.json`). The schema enforces required fields (`id`, `name`, `tactic`, `description`, ~~`severity`~~, ~~`root_cause`~~, `aegis_mitigation`), validates identifier patterns (technique IDs must match `^T[0-9]{4,5}(\.[0-9]{3})?$` (parent and sub-technique forms)`^T[0-9]{4}$`, tactic IDs must match `^TA[0-9]{3}$`), ~~constrains severity values to the enumeration (critical, high, medium, low)~~, and rejects additional properties. Validation can be performed using any JSON Schema Draft 2020-12 compliant validator.

STIX Compliance

The STIX 2.1 Bundle conforms to the OASIS STIX 2.1 specification [?]. All objects include required STIX properties (`type`, `spec_version`, `id`, `created`, `modified`). Attack patterns include kill chain phases linking techniques to their parent tactics. Relationship objects use STIX-standard relationship types (`uses`, `mitigates`, `related-to`).

Cross-Reference Completeness

Every technique has a corresponding entry in the regulatory cross-reference matrix. Each entry includes at minimum one NIST AI RMF function mapping, one EU AI Act article reference, and one OWASP LLM Top 10 category mapping.

Severity Removed in v2.3 ~~Severity-Distribution~~

~~The 20 techniques distribute across severity levels as follows: critical (6), high (9), medium (4), low (1). This distribution reflects the empirical finding that most ungoverned agent behaviors have high or critical impact potential.~~

In v2.3 the severity field has been removed from technique definitions to align with MITRE ATT&CK and ATLAS conventions, which leave contextual scoring to consumers.

Severity is appropriately context-dependent (function of asset, environment, and controls present) rather than a fixed taxonomy attribute.

RECORDS AND STORAGE

File Inventory

Storage Locations

The dataset is permanently stored in three locations:

- 1) **IEEE** **DataPort**—DOI: [pending v2.3 deposit \(v1.0 archived at 10.21227/f87b-1d57\)](https://doi.org/10.21227/f87b-1d57) ~~10.21227/f87b-1d57. Authoritative repository for this submission.~~
- 2) **Zenodo (dataset)**—DOI: [10.5281/zenodo.20171712](https://doi.org/10.5281/zenodo.20171712) (v2.3); prior versions remain at their original DOIs (v1.0: [10.5281/zenodo.19225676](https://doi.org/10.5281/zenodo.19225676); v2.1: [10.5281/zenodo.19251098](https://doi.org/10.5281/zenodo.19251098); v2.2: [10.5281/zenodo.19483999](https://doi.org/10.5281/zenodo.19483999)) ~~10.5281/zenodo.19225676.~~ Open access, indexed by OpenAIRE and Google Dataset Search.
- 3) **Zenodo (source repository)**—DOI: [10.5281/zenodo.19235296](https://doi.org/10.5281/zenodo.19235296) (v1.0 source snapshot retained for the reviewed record; v2.3 source snapshot available at the aegis-governance repository tag `atx-1-v2.3`) ~~10.5281/zenodo.19235296. Archived snapshot of the aegis-governance repository at tag atx-1-v1.0, including all source files, schemas, and documentation.~~

All storage locations serve identical file content under the Apache 2.0 license.

INSIGHTS AND NOTES

Use Cases

Threat intelligence integration. The STIX 2.1 Bundle can be imported directly into SIEM platforms and threat intelligence tools that support STIX, enabling organizations to incorporate agentic AI threat patterns into existing security monitoring workflows.

Governance policy development. The technique-to-mitigation mapping provides a direct link between observed failure modes and architectural countermeasures, supporting evidence-based governance policy authoring for AI agent deployments.

Regulatory compliance mapping. The regulatory cross-reference matrix enables organizations to trace specific agentic AI risks to regulatory requirements across NIST AI RMF, EU AI Act, and OWASP frameworks, supporting compliance documentation and gap analysis.

Research benchmarking. The empirically grounded technique catalog provides a structured basis for evaluating the effectiveness of AI governance frameworks against documented failure modes.

Limitations

TABLE 2. ATX-1 File Inventory

File	Format	Size (bytes)	Description
atx-1-bundle.json	STIX 2.1 JSON	151,05561,795	Complete taxonomy: 10 tactics, 58 attack patterns, 29 mitigations, 106 relationships9-tactics, 20-attack patterns, 20-mitigations, 54-relationships
atx-1-techniques.json	JSON Array	69,09018,578	All 58 entries (29 parent techniques, 29 sub-techniques)All 20-techniques with severity, root cause, case studies, OWASP mapping, AEGIS mitigations
atx-1-regulatory-crossref.json	JSON Object	51,46612,491	Technique mappings to NIST AI RMF, EU AI Act, OWASP LLM Top 10, ATM-1 scenarios
atx-technique.schema.json	JSON Schema	3,3032,459	Validation schema (Draft 2020-12) for technique definitions

ATX-1 v2.3 derives its technique definitions primarily from the *Agents of Chaos* study [?], with v2.1 additions (the four techniques under TA010) derived from RFC-0006 adversarial testing of the AEGIS Claude Code plugin. The core findings are corroborated by independent research on multi-agent vulnerabilities [?], cross-domain security challenges [?], governed system failure modes [?], and additional empirical work cited in the Validation section below. The convergence of multiple independent research groups on the same failure patterns provides confidence in the taxonomy’s coverage of the dominant threat classes. ~~ATX-1 v1.0 derives its technique definitions primarily from the *Agents of Chaos* study [?], with the core findings corroborated by independent research on multi-agent vulnerabilities [?], cross-domain security challenges [?], and governed system failure modes [?]. While the technique catalog may not be exhaustive, the convergence of multiple research groups on the same failure patterns provides confidence in the taxonomy’s coverage of the dominant threat classes.~~ Future versions will incorporate additional empirical sources as the field of agentic AI governance matures.

~~The severity ratings reflect the author’s assessment based on the documented case studies and are not derived from a quantitative risk model. Organizations should calibrate severity ratings to their specific deployment context.~~

Severity ratings are no longer carried by ATX-1 as of v2.3 (see the note under *Severity Distribution* above). Consumers should attach context-specific severity in their own deployments, consistent with MITRE ATT&CK and ATLAS conventions.

Extensibility

The JSON Schema and STIX format support extension. New techniques can be added by conforming to the schema and assigning identifiers in the existing numbering scheme. The STIX Bundle supports standard extension mechanisms for custom properties.

SOURCE CODE AND SCRIPTS

The ATX-1 dataset and all associated schemas are maintained in the AEGIS Governance repository [?]. The source files are located in the docs/atx/ directory under the Apache 2.0 license.

The STIX 2.1 Bundle can be consumed using the stix2 Python library [?] (v3.0+):

```
from stix2 import parse

with open("atx-1-bundle.json") as f:
    bundle = parse(f.read(), allow_custom=True)

for obj in bundle.objects:
    if obj.type == "attack-pattern":
        ext = obj.external_references[0]
        print(f"{ext.external_id}:{obj.name}")
```

Technique validation against the JSON Schema can be performed using ajv-cli:

```
npx ajv validate \
-s atx-technique.schema.json \
-d 'atx-1-techniques.json#/0'
```

No custom software was developed specifically for the dataset. The taxonomy was authored manually based on the empirical analysis described in Collection Methods and Design.

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Kenneth Tannenbaum: conceptualization, taxonomy design, data collection and analysis, machine-readable encoding, manuscript preparation.

This work was done independently of any grants or awards, produced under AEGIS Operations LLC.

The author is the founder of the AEGIS Initiative, which develops the governance architecture that ATX-1 informs. The dataset is published under Apache 2.0 and is designed for use with any governance framework, not exclusively AEGIS.

AI Disclosure: This dataset and manuscript were prepared with the assistance of Claude (Anthropic) as an analysis and writing tool. The taxonomy structure, technique definitions, severity assessments, and all technical content are the author's original work.

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